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A turnstile mechanism for fronts propagating in fluid flows

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We consider the propagation of fronts in a periodically driven flowing medium. It is shown that the progress of fronts in these systems may be mediated by a turnstile mechanism akin to that found in chaotic advection. We first define the modified ("active") turnstile lobes according to the evolution of point sources across a transport boundary. We then show that the lobe boundaries may be constructed from stable and unstable *burning invariant manifolds* (BIMs)—one-way barriers to front propagation analogous to traditional invariant manifolds for passive advection. Because the BIMs are one-dimensional curves in a three-dimensional ($xy\theta$) phase space, their projection into xy-space exhibits several key differences from their advective counterparts: (lobe) areas are not preserved, BIMs may self-intersect, and an intersection between stable and unstable BIMs does not map to another such intersection. These differences must be accommodated in the correct construction of the new turnstile. As an application, we consider a lobe-based treatment protocol for protecting an ocean bay from an invading algae bloom. © *2013 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4824675]

Fronts in flowing media are the combination of two complementary physical phenomena. Front propagation is governed by geometric evolution, built around the structure of rays and perpendicular fronts. Fluid flows fold impurities into sinewy structures reflecting the chaotic nature of advection, even for simple laminar flows. Here, we study the interplay of these two dynamics. A central concept in both front propagation and fluid flows is that of *transport*—moving something from here to there. A laser pulse traveling through fiber optic cable transports information, while an ocean wave transports energy and surfers. While some instances of fluid transport have a similar linear character, e.g., the oceanic conveyor belt that moves heat around the globe, we will be more interested in fluid transport as a means of mixing, a process that depends on the chaotic nature of the flow. This study examines how chaotic transport in fluids is augmented by propagation. Our approach draws from the existing geometric theory of transport in fluid flows known as the *turnstile mechanism*. This mechanism is a first-order description of chaotic mixing, which describes the exchange of fluid packets between two adjacent regions. We develop a reformulation of this theory that naturally encompasses the addition of front propagation.

I. INTRODUCTION

We build upon the foundation subjects of fluid flow and front propagation, both central dynamical mechanisms in Nature. Exemplars of fluid flow include oceanic and atmospheric flow, industrial mixing, and microfluidics. We might also include such systems as phase space flow and swarms of moving agents. Well-studied cases of front propagation include optical and acoustic wavefronts, phase transition fronts, and chemical reaction fronts. Again we can broaden our view to include growth of algae blooms, spread of infectious disease, and dissemination of information and beliefs.

It is often the case that the substrate supporting the propagation of a front is not static, and certainly may be some form of flowing fluid. The combination of these two elemental dynamics yields a variety of new and interesting phenomena while maintaining features of each component.

Recent experiments have combined chemical reaction front propagation with laboratory scale flows.[1–5] The flows were generated using magneto-hydrodynamic forcing in a quasi-two-dimensional fluid. The reaction fronts were generated through Belousov-Zhabotinsky chemistry. A variety of phenomena were investigated including mode-locking and front-pinning. Direct numerical partial differential equation (PDE) simulations[6,7] have supported these experiments.

Here, we take a dynamical systems perspective. Invariant manifolds are key to understanding passive advection in time-independent and time-periodic fluid flows.[8] Recent experiment and theory[3,9,10] have shown that these invariant manifolds survive, but are modified by the addition of front propagation to the dynamics. The original (advective) invariant manifolds undergo a bifurcation whereby they split into two nearby *oriented* invariant manifolds, called *burning invariant manifolds* (BIMs). Such manifolds are oriented in that they are *one-sided* barriers to the progress of fronts in fluid flows.

The fundamental geometric construct for understanding chaotic transport in passive flows is the turnstile mechanism.[11] The turnstile is built from segments of advective stable and unstable invariant manifolds. Since BIMs are relevant for front propagation, this suggests that a turnstile-type mechanism may also be at work in active fluid flows.

We begin in Sec. II with a motivating example—an oceanic algae bloom invades a bay. In Sec. III, we offer a review of the turnstile mechanism in passive advection. Then, Sec. IV presents the dynamical system for fronts in flows. Section V defines the new burning lobes in terms of point stimulations. We then recall the necessary features of BIMs from previous works (Sec. VI). Sections VII and VIII examine the

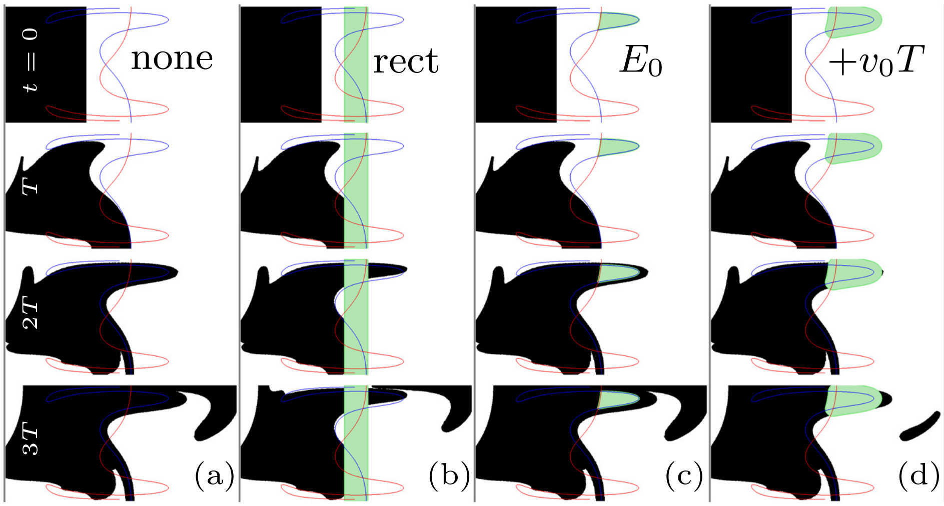

FIG. 1. Initial algae population (black) impinges from the left. Algae evolved using a 400 × 800 grid-based algorithm as in Ref. 6. Stable (red) and unstable (blue) advective invariant manifolds are overlaid. Time increases down. (a) No suppression, algae invades. (b) Large rectangular suppression (green) around undriven separatrix; the algae still invades. (c) Lobe-based suppression using E_0 lobe; in only one cycle, treated area is enveloped and made irrelevant. (d) The E_0 lobe "fattened" by v_0T still does not prevent invasion. Treated areas: $|\text{rect}| = 0.2$, $|E_0| = 0.035$, $|E_0 + v_0T| = 0.101$. Channel flow given by Eq. (1) ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

modification to advective turnstiles necessary for our new turnstile given small and large front propagation speeds, respectively. We end by returning to the algae example and discuss an efficient treatment based on the developed burning turnstile (Sec. IX).

II. APPLICATION: ALGAE INVASION

There is growing interest in the behavior of biological and chemical systems in fluid flows.[12,13] For instance, recent studies have examined the interplay between rapid population growth in plankton and the underlying ocean currents.[14] Here, we describe a scenario in which knowledge of the burning turnstile (developed in this paper) can be used to more efficiently mitigate the impact of an impinging front. Imagine an ocean bay that we wish to protect from an invading algal bloom. Suppose that the surface ocean flow is well-modeled by a 2D flow.[15] A separatrix may be generated by an ocean current parallel to the coastline and across the mouth of the bay. Additionally, diurnal tides are a reasonable source of periodic perturbation producing a "chaotic turnstile" mechanism for ocean water to enter and exit the bay.

Suppose the goal is to protect the bay by applying an algaecide to the surface of the water in synchrony with the tide. Furthermore, imagine that this substance is 100% effective, but is active for only a short time, possibly due to photo-degradation. Importantly, treatment is often costly or toxic to other ocean life, so we wish to apply the minimum amount necessary. We do not presume the details accurately reflect a real-life scenario. The intention is to illustrate a plausible motivation for suppressing the invasion of a front. We use a simple channel flow model given by Eqs. (1).

Figure 1 shows the algae invasion under different treatment protocols. The first column shows the evolution of algae with no treatment. The algae quickly invades the right-hand-side. Notice also the way in which it invades. The rectangular initial condition quickly develops a correlation with the advective invariant manifolds (shown in blue and red).

In the first suppression protocol (Fig. 1(b)), we apply a rectangular strip of algaecide spanning the mouth of the bay. While this protocol is simple to implement, it does not take advantage of our understanding of the dynamical structure, and as a result either fails to prevent the algae invasion, or must use an excess of suppressant.

A better strategy attempts to make use of oceanic flow structures. In particular, one could focus treatment on the lobes responsible for left-to-right transport. In the third column, we specifically target the E_0 lobe. At $t = 2T$, it is clear that this treatment has missed a significant portion of the invading algae. In the final step $t = 3T$, we see that the treated area has been completely filled in, and we are actually no better off than in the untreated case.

Since these lobes are related to passive transport and the algae are expanding in addition to flowing, a natural strategy (Fig. 1(d)) is to enlarge the passive lobes by v_0T—the distance traveled by the algae front in quiescent fluid over one period.[16] At $t = 2T$, this protocol appears to be almost successful. There is only a small amount of algae that has grown beyond the tip of the treated area. Then, at $t = 3T$, when the bulk of the population has advanced slightly, the amount of new algae that penetrates is significant. In addition, the small amount from before has now grown into a sizable blob. At best, this extended lobe protocol may delay the full-blown invasion for a few forcing cycles.

While the last protocol is correct in spirit (it takes into account both flow and propagation), it wrongly treats these dynamics as independent and so misses a crucial feedback—when a front propagates forward, it enters a new region and thus experiences a new flow. This continuous feedback must be taken into account.

The rest of this paper develops the details of the "correct" protocol by generalizing the turnstile to accommodate front propagation. A reader interested in the application of this new protocol to the algae model may wish to skip ahead to Sec. IX.

III. ADVECTIVE TURNSTILE REVIEW

Time-independent, two-dimensional, incompressible flows are well characterized by the structure of their fixed points and separatrices (Fig. 2(a)). The separatrices are invariant curves under the flow. They divide the fluid trajectories into classes with qualitatively distinct behaviors. Unlike the invariant tori that are also present in these flows,

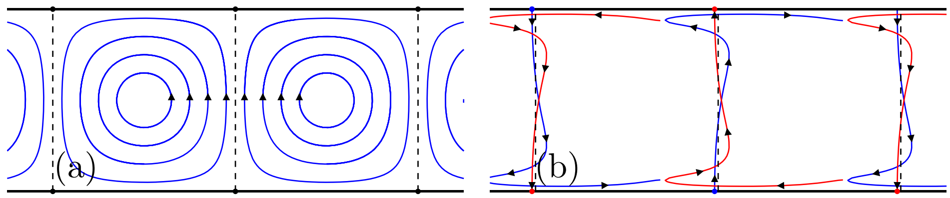

FIG. 2. Structure in alternating-vortex-chain fluid flow. (a) Time-independent flow forms vortex cells bounded by separatrices (dashed lines). ($b = 0, \omega = 2\pi, \phi = 0$) (b) Periodic lateral perturbation breaks each separatrix into stable (red) and unstable (blue) manifolds. ($b = 0.1, \omega = 2\pi, \phi = 0$).

separatrices divide trajectories that exponentially diverge or converge.

The introduction of a time-periodic perturbation leads to the splitting of each separatrix into distinct stable and unstable invariant manifolds (Fig. 2(b)). These manifolds are not invariant with respect to the time-varying flow, but rather the map $M_a : \mathbb{R}^2 \to \mathbb{R}^2$ obtained by integrating the flow over one forcing cycle. (The subscript a is for "advective" as opposed to b for "burning" later.)

As a model system for illustrating these structures, we use the alternating vortex chain (AVC) in a channel geometry,[17]

$$\begin{aligned} u_x(x, y, t) &= +\sin(\pi[x + b\sin(\omega t - \phi)])\cos(\pi y), \\ u_y(x, y, t) &= -\cos(\pi[x + b\sin(\omega t - \phi)])\sin(\pi y), \end{aligned} \tag{1}$$

where $0 \le y \le 1$, and b, ω, and ϕ are dimensionless parameters that describe the periodic forcing amplitude, frequency, and phase, respectively. (We use the phase ϕ to choose a particular Poincaré section.) This model has free-slip boundary conditions (BCs). We will use this same model in following sections when discussing active fluids.

For small perturbations and short times, the unforced separatrices are still fairly good classifiers of particle trajectories. However, these separatrices are now traversed by certain trajectories. This is sometimes referred to as a "leaky separatrix." Importantly, this is not "leakiness" in the sense of diffusion. Rather, there is structure to the leakiness. One iteration of the map causes a localized packet of fluid to be transported across from left to right, and a corresponding packet from right to left.

A. Set definition of lobes

To formalize these ideas, we first choose a transport boundary (TB) that divides the fluid in two allowing us to define a flux. In this paper, we focus on flow in a channel geometry and choose a transport boundary that defines left- and right-hand sides (L, R). (Starting in Sec. V, we specialize to transport from left to right.) Given this transport boundary, we define lobes as sets of points that cross this boundary from left to right, or right to left, upon one iteration of the map M_a. The two sets, before and after the map, are referred to as lobes. The escape lobes E_i and capture lobes C_i are defined as

$$\begin{aligned} E_0 &= \{p \in R : \exists q \in L, M_a(q) = p\}, \\ E_{-1} &= \{p \in L : M_a(p) \in R\}, \\ C_0 &= \{p \in L : \exists q \in R, M_a(q) = p\}, \\ C_{-1} &= \{p \in R : M_a(p) \in L\}. \end{aligned} \tag{2}$$

We use the same notation to denote the map applied to a set of points $M_a(A) = \{M_a(p) : p \in A\}$. Using this notation and the invertibility of M_a, it is somewhat more intuitive to write

$$\begin{aligned} E_0 &= R \cap M_a(L), \\ E_{-1} &= M_a^{-1}(R) \cap L, \\ C_0 &= L \cap M_a(R), \\ C_{-1} &= M_a^{-1}(L) \cap R, \end{aligned} \tag{3}$$

noting that $E_0 = M_a(E_{-1})$ and $C_0 = M_a(C_{-1})$.

In Figure 3, we choose the advective ($b = 0$) separatrix (vertical line in the middle) as our transport boundary. Examining the action of one map iteration, we find eight regions of interest. Two disjoint regions map to two other disjoint regions going left to right ($E_{-1} \to E_0$), and similarly going right to left ($C_{-1} \to C_0$). Notice that these regions are bounded both by segments of the separatrix and also by segments of its forward and reverse iterates, which do not coincide with the separatrix. Also, note the overlapping of regions and the amount of total area involved. While this analysis of single-step transport is correct, it can be simplified by a different choice of transport boundary.

B. Transport boundary—invariant manifolds

While the above definition yields well-defined lobes given *any* transport boundary, using segments of stable and unstable invariant manifolds yields a boundary with a (locally) minimal flux.[8,11] In doing so, we describe the transport process using "native" structures. Two hyperbolic fixed

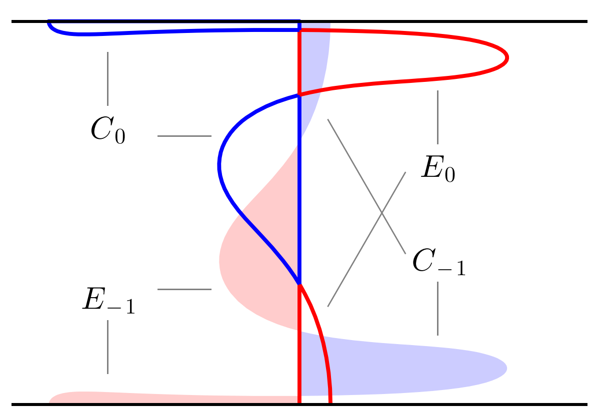

FIG. 3. Separatrix (central vertical line) is chosen as transport boundary for periodically forced AVC. Left to right transport: red filled regions map to red outlined regions ($E_{-1} \to E_0$). Right to left transport: blue filled regions map to blue outlined regions ($C_{-1} \to C_0$). ($b = 0.4, \omega = 5, \phi = \pi$).

points (on opposite sides of the channel) are denoted z^A and z^B. The stable and unstable manifolds attached to these points are denoted W^S and W^U, respectively. The finite segments connecting these fixed points to the intersection point p_0 are denoted $W^S[z^A, p_0]$ and $W^U[z^B, p_0]$. The intersection point is required to be a *primary intersection point* (pip);[18] that is, $W^S[z^A, p_0] \cap W^U[z^B, p_0] \setminus \{z^A, z^B, p_0\} = \emptyset$. The transport boundary is then defined as $W^S[z^A, p_0] \cup W^U[z^B, p_0]$.

In Fig. 4, we illustrate the lobes derived from this new transport boundary. Importantly, these lobe boundaries are comprised solely of segments of these same invariant manifolds. Each lobe is also composed of a smaller number (here just one) of disjoint regions having smaller total area.[11]

The C_{-1} and E_{-1} lobes share a point in their boundary and, loosely speaking, the complementary transport of these packets into C_0 and E_0 corresponds to a rotation (and shift) about this point. Consequently, this exchange is referred to as a *turnstile mechanism*.[11,19] Generalizing the connection between the evolution of sets and bounding invariant manifolds to the case of front propagation is one of the main results of this paper.

IV. A DYNAMICAL SYSTEM FOR FRONTS IN FLOWS

A. Assumptions

We assume that the two-dimensional fluid is partitioned into regions that can be identified as either "burned" or "unburned"—we use "burned" generically to refer to the region bounded by an outward propagating front. That is, these regions are separated by well-defined one-dimensional (codimension one) curves or *fronts*. This is known as the "thin (sharp) front," or "geometric optics" regime.[20] For instance, if our front is an autocatalytic chemical reaction front, this limit requires that the diffusion rate is small compared with the reaction rate.[6]

We assume that each front element (infinitessimal piece of the one-dimensional front) propagates perpendicular to the front at a speed v_0 measured in the comoving fluid frame, thereby expanding the burned region. To be clear, we require that this speed v_0 be isotropic, homogeneous, and

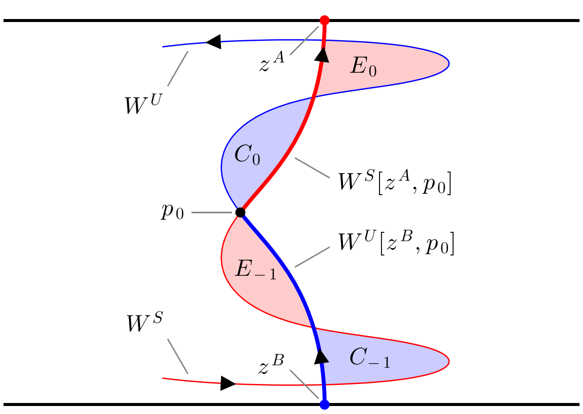

FIG. 4. Turnstile mechanism in periodically forced alternating vortex chain. Segments of stable (red) and unstable (blue) manifold are combined to form transport boundary (bold). Shaded areas indicate the two pathways for crossing this transport boundary. Use of dynamically generated transport boundary yields simpler lobe structure. ($b = 0.4, \omega = 5, \phi = \pi$).

time-independent. Note that we have implicitly assumed that the front elements are non-interacting (fronts do not change their behavior when faced with a collision), and their motion does not depend on the front curvature.[13,21]

Finally, we assume that there is no effect of the front propagation on the underlying fluid flow. In the case of a chemical reaction front, this means that the reaction is not sufficient to change the fluid density, viscosity, etc. This is a good approximation in such experiments as Refs. 1–3.

B. Front element ordinary differential equation (ODE)

These assumptions lead to Eqs. (4), a three-dimensional ODE governing the evolution of each front element. The total translational motion of a front element is the vector sum of the fluid velocity and the front propagation velocity in the fluid frame (Eq. (4a)). The change in orientation is determined entirely geometrically; Eq. (4b) describes the angular velocity of a material line embedded in the fluid

$$\dot{\mathbf{r}} = \mathbf{u} + v_0\hat{\mathbf{n}}, \tag{4a}$$

$$\dot{\theta} = -\hat{\mathbf{n}}_i \mathbf{u}_{i,j} \hat{\mathbf{g}}_j. \tag{4b}$$

The variables $\mathbf{r}$ and θ denote the position and orientation of each front element. The prescribed fluid velocity is given by $\mathbf{u}$. Again, v_0 is the front propagation speed in the comoving fluid frame. For compactness, we use the variables $\hat{\mathbf{g}} = (\cos\theta, \sin\theta)$ and $\hat{\mathbf{n}} = (\sin\theta, -\cos\theta)$ to indicate the tangent to the front element and the normal direction (propagation direction), respectively. Finally, $\mathbf{u}_{i,j} = \partial u_i/\partial r_j$ and repeated indices are summed.

We define $M_b : \mathbb{R}^2 \times S^1 \to \mathbb{R}^2 \times S^1$, the evolution of front elements, by integrating Eq. (4) for one forcing period. The map M_b is the "burning" extension of M_a.

For any curve $C = (x(\lambda), y(\lambda), \theta(\lambda))$, we denote its iterate by $M_b(C)$. We compute $M_b(C)$ numerically by integrating Eq. (4) for each point along the approximated curve C. Because of the potential for very large stretching in this map, front evolution and invariant manifold construction require the insertion of interpolated points to maintain sufficient density.[22]

It is important to note that not all curves in this 3D phase space correspond to fronts. Let C be parameterized by the xy Euclidean length along the curve. Then, the *front-compatibility criterion* requires that $\theta(\lambda)$ everywhere equals the angle θ' describing the local tangent to the curve's xy-projection, $\tan\theta'(\lambda) = (dy/d\lambda)/(dx/d\lambda)$.

Any parameterized curve in xy-space lifts to a unique curve in $xy\theta$-space, which satisfies the front-compatibility-criterion. In particular, the burned region A has oriented boundary ∂A, which lifts in this way. We shall use the term *front* for both the curve in xy-space and its lift to $xy\theta$-space.

The burned region itself evolves under the map $\mathcal{M} : 2^{\mathbb{R}^2} \to 2^{\mathbb{R}^2}$. This map can be defined in terms of its action on "point stimulations" ($\delta(x)$). The map of a region A is simply the union of images of each point in A, $\mathcal{M}(A) = \cup\{\mathcal{M}(p) : p \in A\}$. The intuition is simply that if stimulation p_1 evolves to region R_1 and p_2 to R_2, then stimulating both p_1 and p_2 will result in the reacted domain $R_1 \cup R_2$. Note that $\mathcal{M}$ acting on sets is not one-to-one, i.e., two distinct burned regions can evolve forward to the same burned region. This differs from the passive case in which M_a acting on sets was invertible (Sec. III).

It is worth pointing out the connection to the G-equation[23] and the Fisher-Kolmogorov-Petrovskii-Piskunov (FKPP) equation.[24,25] The G-equation is a PDE whose level set $G = 0$ represents the evolving front. It can be shown that this is equivalent to Eq. (4).[26] The FKPP equation is a PDE that describes the evolution of a continuous valued "reactant field" under advection, reaction, and diffusion. Importantly, it reduces to the G-equation in the sharp front limit, where each fluid parcel is either completely burned or unburned. By representing front evolution with Eq. (4), we are able to apply the tools of low-dimensional dynamical systems theory.

V. BURNING TURNSTILE: "BURNSTILE"

Here, we define the turnstile relevant for active fluids in terms of the behavior of sets, much like was done for the advective turnstile (Eqs. (2)). The essential differences arise due to the everywhere-expanding property of the map $\mathcal{M}$.

Because of the asymmetry of the map $\mathcal{M}$ (unburned fluid can become burned but not vice versa), there exists an asymmetry in the lobe definitions depending on whether the burned region is to the left or the right of the transport boundary. Importantly, this choice imbues the transport boundary with an orientation, or "propagation direction." For the remainder of this paper, we consider left-to-right propagation ($TB = \partial L$). The right-to-left case is defined analogously. Note, however, that while the advective turnstile is the same for transport in either direction, there is actually a separate burning turnstile for right-to-left propagation.

A. Escape

We begin with the escape process. Since advection maps points to points, any given point begins and ends either right or left of the transport boundary, and so crossing (e.g., escaping) is well defined. However with the addition of front propagation, points map to regions under $\mathcal{M}$ and so a choice arises; is a point stimulation, which immediately gives rise to a burned region, said to escape when *all* of the region crosses the transport boundary? Or instead when *any* of the region crosses? Looking to physical examples such as autocatalytic chemical reactions, fires, algae blooms, infections, etc., it seems reasonable that the more interesting definition is when *any* crosses; we are generally not concerned whether an *entire* wildfire crosses the firebreak. This motivates the following definitions of escape lobes in the active system (compare Eqs. (2)):

$$\begin{aligned} E_{-1} &= \{p \in L : \mathcal{M}(p) \cap R \neq \emptyset\}, \\ E_0 &= \{p \in R : \exists p' \in L, p \in \mathcal{M}(p')\}. \end{aligned} \tag{5}$$

The escape lobe E_{-1} is the set of all points left of the transport boundary such that the evolution of each point stimulation under $\mathcal{M}$ has some intersection with the area right of the transport boundary. The escape lobe E_0 is the set of all points

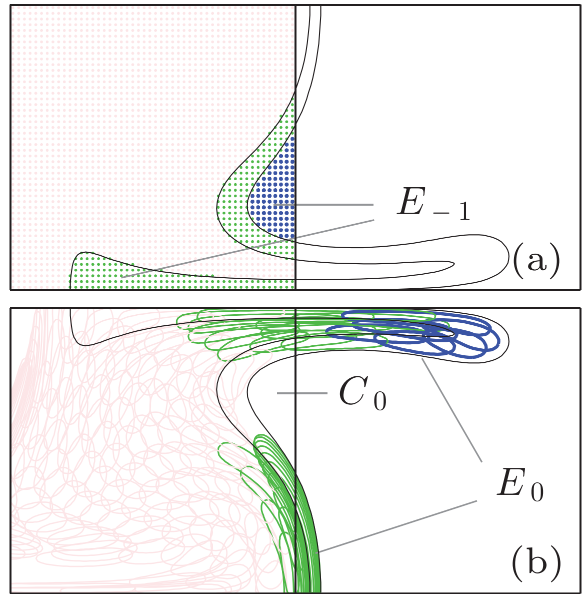

FIG. 5. Vertical line in center ($b = 0$ separatrix) is chosen as transport boundary. Many stimulations are initialized in L (a) and each is evolved forward for time T (b). Each stimulation is then colored according to its subsequent intersection with R: blue (darkest) means completely in R, green (medium) means only partial intersection, and red (light) means no intersection. E_{-1} lobe is blue and green together. E_0 is only that blue and green in R. C_0 is empty in L. For clarity in (b), only 1 in 16 iterates is displayed. Two black curves on left and right are (a) $M_b^{-1}(\partial L)$ and $M_b^{-1}(\partial R)$; (b) $M_b(\partial R)$ and $M_b(\partial L)$ ($v_0 = 0.05, b = 0.3, \omega = 5, \phi = \pi$).

right of the transport boundary that may be reached in one iterate by some point stimulation on the left. See Fig. 5.

B. Capture

The capture process is slightly subtler due to the asymmetry in the burning process (compare Eqs. (2)),

$$\begin{aligned} C_{-1} &= \{p \in R : \mathcal{M}(p) \cap R = \emptyset\}, \\ C_0 &= \{p \in L : \forall p' \in L, p \notin \mathcal{M}(p')\}. \end{aligned} \tag{6}$$

The capture lobe C_{-1} is the set of all points right of the boundary such that the evolution of this point stimulation under $\mathcal{M}$ lies completely on the left. In other words, C_{-1} contains all those stimulations on the right that do not contribute to the subsequent burning of R. The capture lobe C_0 is the set of all points on the left not reachable in one period by any stimulation on the left. Since all points are reachable by some stimulation, C_0 is equivalently those reachable only by a stimulation on the right. See Fig. 6.

We reiterate that, just as in the passive case, these set-based definitions of lobes are well-defined with respect to an arbitrary transport boundary. However, we again wish to make use of the dynamic structure of the system by choosing the transport boundary in a natural way.

C. Recast burning lobe definitions

The burning lobe definitions are phrased in terms of mapping properties of point stimulations. Experimentally, this definition makes good sense because a point's lobe membership is based on the results of an elementary

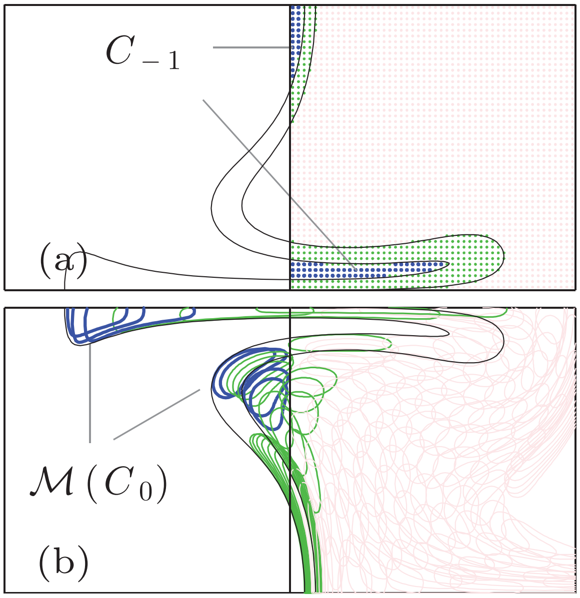

FIG. 6. Many stimulations in R (a) evolved forward for time T (b). Stimulations are then colored according to their intersection with L: blue (darkest) means completely in L, green (medium) means only partial intersection, red (light) means no intersection. Only those that map entirely into L are part of C_{-1} lobe (blue). ($v_0 = 0.05, b = 0.3, \omega = 5, \phi = \pi$).

experiment—the stimulation of just this point (or as close to it as possible).

However, making use of the extension of $\mathcal{M}$ to regions, we can rephrase the definition of the burning lobes. The first two, E_0 and C_0, are relatively intuitive,

$$\begin{aligned} E_0 &= R \cap \mathcal{M}(L), \\ C_0 &= L \cap \mathcal{M}(L)^c, \end{aligned} \tag{7}$$

where A^c indicates the set complement of A.

In order to similarly describe the other two lobes, we need a map inverse. Since $\mathcal{M}$ is many-to-one, we must *choose* an inverse.

Given a final burned region B, there are an infinite number of initial regions A for which $\mathcal{M}(A) = B$. Two useful senses of inverse are derived from considering extreme cases. We define the *loose* and *tight* preimages, respectively, as[27,28]

$$\begin{aligned} \overline{\mathcal{M}}^{-1}(A) &\equiv \{p : \mathcal{M}(p) \cap A \neq \emptyset\}, \\ \underline{\mathcal{M}}^{-1}(A) &\equiv \{p : \mathcal{M}(p) \subset A\}. \end{aligned} \tag{8}$$

The loose preimage is the set of all stimulations that intersect the target, while the tight preimage is the set of only those stimulations that map entirely into the target. It follows that the tight preimage is a subset of the loose one, $\underline{\mathcal{M}}^{-1}(A) \subset \overline{\mathcal{M}}^{-1}(A)$. (This motivates the overline versus underline notation.) Given any A, the loose preimage always has non-zero area while the tight preimage may be empty. Note that $\overline{\mathcal{M}}^{-1}(A)$ and $\underline{\mathcal{M}}^{-1}(A^c)$ form a disjoint partition of the entire space.

We may now write

$$E_{-1} = \overline{\mathcal{M}}^{-1}(R) \cap L, \quad C_{-1} = \underline{\mathcal{M}}^{-1}(L) \cap R. \tag{9}$$

VI. BIM BASICS

We now review the necessary features of burning invariant manifolds.[9,10] While incompressible flows have only hyperbolic (SU) and elliptic (SS) fixed points, all fixed-point stability types (SSS, SSU, SUU, and UUU) are possible in front element dynamics (Eq. (4)). For the purposes of this paper, we will concern ourselves only with the SUU and SSU stability types, and only with invariant manifolds of their singleton dimensions. We will refer to these one-dimensional objects as stable and unstable BIMs, respectively. Previous theoretical and experimental studies have demonstrated that unstable BIMs are one-sided barriers to front propagation in fluid flows, similar to traditional advective invariant manifolds (Fig. 7).

To compute an unstable BIM, we begin with a small front segment that starts at a fixed point of Eq. (4) and extends a small distance in the direction of the unstable eigenvector. This segment is evolved forward under M_b and interpolation points are added[22] to maintain a sufficient density. Stable BIMs are computed analogously using M_b^{-1}.

While in many ways BIMs function like their passive counterparts, there are some important differences. While advective invariant manifolds are codimension-1, BIMs are codimension-2 since they exist in a three-dimensional phase space. (Note that the figures in this work portray *projections* of BIMs into xy-space.) It is then not obvious how BIMs can function as barriers.

It has been shown that BIMs satisfy the front-compatibility criterion and so are (locally) fronts.[29] As such, BIMs are also subject to the *no-passing lemma*—no front can overtake another front from behind. When the leading front is an invariant one (a BIM), the other is bounded by it. Because unstable BIMs are transversely stable not only do they bound fronts but fronts also converge to them making unstable BIMs evident in observations of the dynamics. A similar effect is observed in passive flows.[30]

Just as fronts have a burning direction, so do BIMs. A front near a BIM with the same burning direction will not pass through. Conversely, a front with burning direction opposite a BIM can pass through with no singularity in its motion. This is why BIMs are referred to as *one-sided* barriers.

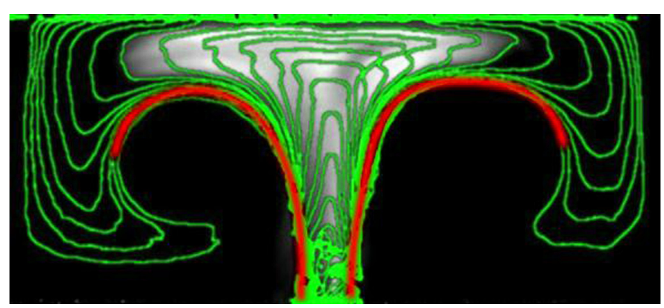

FIG. 7. A chemical reaction front is stimulated at the bottom of the time-independent AVC near a hyperbolic fixed point. The reacted region is advected upward while growing in size. Edge detection applied to this experimental image sequence results in the sequence of green contours. We deduce the presence of BIMs (red) bounding the front progress. (Reprinted with permission from J. Mahoney *et al.*, Europhys. Lett. **98**, 44005 (2012). Copyright © 2012 EPLA.

In the full $xy\theta$-space, BIMs are smooth, non-intersecting curves. However, when projected into the xy-plane, BIMs have cusps and self-intersections. These cusps are physically meaningful in that fronts can wrap around them. Therefore, a BIM loses its bounding behavior at a cusp.

Having established that BIMs are barriers to front propagation, just as advective invariant manifolds are barriers to fluid transport, it might seem that the analogy is complete and all that remains is to construct an active transport boundary from BIM segments. However, in the next few sections, we illustrate several new features of the active system that require our consideration. We proceed by first analyzing the $v_0 \ll 1$ regime.

VII. SMALL v_0: NO NEW INTERSECTIONS

Consider finite segments of a generic advective invariant manifold tangle. Then, there exists a v_0 small enough such that taking the advective manifolds into the left- and right-burning BIMs by a continuous deformation generates no new intersections in these finite segments (see Fig. 8). Examining this simplest topological case reveals two important features in the composition of the active transport boundary.

A. Co-orientation

As v_0 increases from 0, each stable and unstable advective invariant manifold splits into nearby left- and right-burning BIMs (Fig. 8). In joining segments of these BIMs to form a transport boundary, each offers a choice of left- or right-burning. Figure 9 illustrates these four choices. In case (a), the BIM segments are both rightward burning, corresponding to a burned region impinging from the left. Case (c) corresponds to the boundary of a region on the right. Because in each of these two cases, the two BIM segments agree on a burning direction, we call these configurations of joined segments *co-oriented*. In cases (b) and (d), the BIM segments chosen are not co-oriented, and do not correspond to the oriented boundary of a burned L or R.

More precisely, consider the two front elements involved in the intersection and their respective directions of propagation $\hat{\mathbf{n}}_i$. For each, denote by $\hat{\mathbf{t}}_i$ the direction tangent

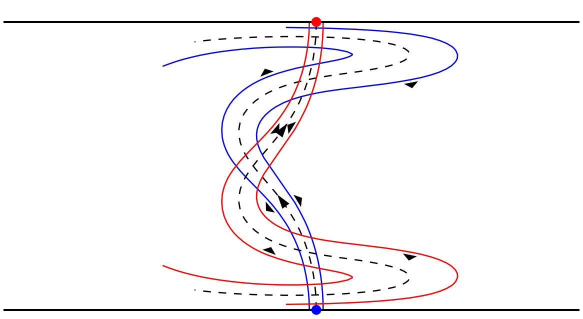

FIG. 8. Each advective manifold (dashed) splits into two oppositely burning BIMs (direction denoted by wide arrows). Stable (unstable) manifolds are attached to the upper (lower) wall. Note the unstable BIMs burn away from the advective manifold while the stable BIMs burn toward it. For small v_0, BIMs follow near the advective manifolds for a long length. $(v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi)$.

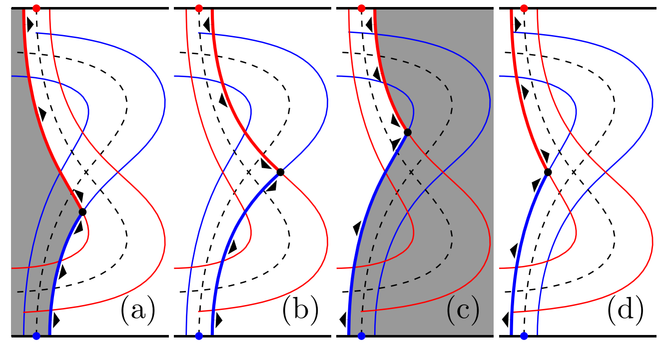

FIG. 9. For cases (a) and (c), the two BIMs agree on a burning direction and so are *co-oriented*. In cases (b) and (d), the two BIMs are mismatched. $(v_0 = 0.1, b = 0.3, \omega = 5, \phi = 0)$.

to the corresponding BIM segment and oriented away from the intersection. If the two cross-products $\hat{\mathbf{n}}_i \times \hat{\mathbf{t}}_i$ are of opposite sign, the segments are co-oriented.

If we were to choose a non-co-oriented configuration (say case (b)), then a burned region abutting this curve from the left (right) will burn through the stable (unstable) BIM segment immediately. Consequently, neither this BIM segment, nor its iterate would bound the iterated burned region. Therefore, our first new requirement of the active transport boundary is that it be composed of BIM segments that are co-oriented. For the remainder of this paper, we analyze fronts progressing from left to right as in Fig. 9(a).

B. Intersections: Splitting, continuation, connection, concavity, convexity

In the advective turnstile, an intersection between stable and unstable invariant manifolds must map to another such intersection. This fact allows us to neatly equate the boundary segments of a lobe iterate with the iterates of the initial boundary segments.

In the active system, an intersection in the xy-projection of two BIMs is not an intersection in $xy\theta$-space. Generically, the BIM projections meet with different orientations and so two *different* front elements participate in their xy-intersection. More importantly, these two front elements evolve forward splitting apart in xy-space (Fig. 10(a)). This might seem to foil an active generalization of the turnstile. However, the bounding property of both advective and burning invariant manifolds is not a pointwise property, but a property of curve segments. This implies that the functional intersection is not necessarily lost; we can define an *intersection continuation* by tracking the continuous evolution of the intersection of the two BIMs. Figure 10(a) shows that while the two front elements initially involved in the intersection always separate, a natural continuation of this intersection can be defined. This continuation outruns each of the two front elements by a factor of $1/\cos\alpha$ in the comoving fluid frame.

Figure 10(a) is compatible with two different burned regions. In Fig. 10(b), the burned region has a *concave corner* while the region in Fig. 10(c) has a *convex corner* (compare Figs. 9(a) and 9(c)). In order to understand the evolution of these burned regions using Eqs. (4) for front

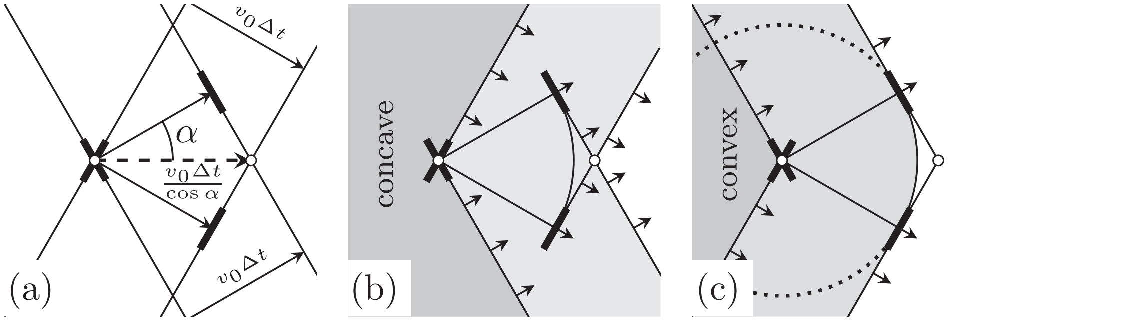

FIG. 10. (a) Two fronts intersect. As they propagate forward, their intersection moves faster than v_0. The initially intersecting front elements are now separated. (b) Concave corner. The front elements initially at the intersection become enveloped in the reacted region. The intersection continuation remains physically relevant. (c) Convex corner. Front elements at intersection remain on the physical boundary. The intersection continuation runs ahead of the reacted region leaving an unburned swallowtail shape behind. The arc is not the evolution of either front segment.

propagation, we need to properly join the two original front segments at the corner. Remembering that fronts are curves in $xy\theta$-space, it is clear that the two front segments alone form a discontinuous curve. We join these two front segments with a *connection* $\mathcal{C}$ that is simply a straight line connecting the θ values of the two front elements at the corner. Since $\theta \in S^1$, there are two distinct intervals between any two θ values; the physically correct choice is the shorter of the two intervals. The two segments joined with the connection are now a continuous front in $xy\theta$-space.

In Fig. 10(b), the connection evolves into the arc of a circle and is immediately burned past by both front segments. Further, the front segments partially overtake each other sending finite portions into the reacted region. The boundary of the final reacted region $\mathcal{M}(R)$ is a subset of the evolved front, $\partial\mathcal{M}(R) \subsetneq M_b(\partial R)$.

In Fig. 10(c), the connection unfolds in the same way, but it remains on the boundary of the reacted region, $\partial\mathcal{M}(R) = M_b(\partial R)$. This connection, functioning like a point stimulation made at the convex corner, is a boundary between burned and unburned that is not the iterate of, or continuation of, either original front segment; it generates "new boundary." Also notice that the intersection continuation outruns this point stimulation. This concept will lead to the second requirement of the active transport boundary.

C. Pips: Splitting and continuation

Having now described front element splitting and intersection continuation and the effect of concavity on the physical relevance of the connection, we apply this to pips between the BIMs. First, we establish some notation for pips, comparing the passive and active cases.

Recall that for advective dynamics, a point p is an *advective pip* if

$$W^S[z^A, p] \cap W^U[z^B, p] \setminus \{z^A, z^B, p\} = \emptyset \tag{10}$$

(see Fig. 4). The image and preimage of an advective pip p_0 are denoted $M_a(p_0) = p_1$ and $M_a^{-1}(p_0) = p_{-1}$, respectively. For any advective pip p_k, $M_a^n(p_k) = p_{k+n}, n, k \in \mathbb{Z}$.

We now consider pips in the active case. The tuple of two intersecting front elements $p \equiv (p^S, p^U)$, where $p^S, p^U \in \mathbb{R}^2 \times S^1$, is a *burning pip*, if

$$\Pi(W^S[z^A, p^S]) \cap \Pi(W^U[z^B, p^U]) \setminus \Pi(\{z^A, z^B, p^S, p^U\}) = \emptyset, \tag{11}$$

where Π denotes projection into the xy-plane. In Fig. 11, we have labeled the burning pips p_i and q_i (solid dots). We write p_i^S (p_i^U) to indicate the stable (unstable) component of the ith burning pip $p_i \equiv (p_i^S, p_i^U)$, and similarly for q_i. Since a burning pip p_0 splits, $M_b(p_0^S) \neq p_1^S$, $M_b(p_0^U) \neq p_1^U$, $M_b^{-1}(p_0^S) \neq p_{-1}^S$, and $M_b^{-1}(p_0^U) \neq p_{-1}^U$. Generically, $M_b^n(p_k) \neq p_{k+n}, \forall n, k \in \mathbb{Z}, n \neq 0$. However, continuation of intersections takes $p_i \to p_{i+1}$ and $q_i \to q_{i+1}$. In Fig. 11, unfilled dots denote iterates of the two components of each burning pip (solid dots) along either the stable or unstable BIM.

For instance, examine the evolution of p_{-1} in Fig. 11. Because of front element splitting, $M_b(p_{-1}) \neq p_0$. Instead, the two front elements p_{-1}^S and p_{-1}^U that compose p_{-1} map to $M_b(p_{-1}^S)$ and $M_b(p_{-1}^U)$. Notice that the two points $M_b(p_{-1}^S)$ and $M_b(p_{-1}^U)$ have mapped, respectively, behind the local unstable and stable BIMs at p_0.

Unlike the passive case in which the boundary of a lobe exactly maps to the boundary of the iterated lobe, in the active case, the stable (unstable) boundary segment of a lobe maps to a stable (unstable) segment that is either shorter or

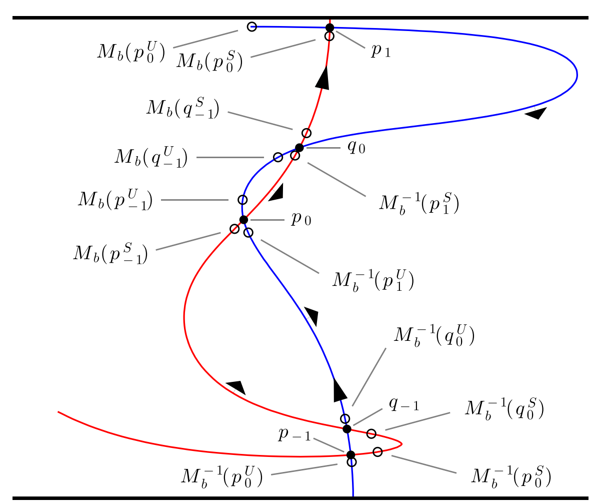

FIG. 11. Demonstration of the splitting of intersection points. Intersection continuations p_i and q_i (solid dots) represent two coincident front elements. These elements are mapped forward and backward (open dots). ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

longer than the stable (unstable) boundary of the subsequent lobe. We use this knowledge, and the fact that pips alternate between concave and convex, to derive the following relations:

$$\begin{aligned} M_b(W^S[p_{-1}, q_{-1}]) &= W^S[M_b(p^S_{-1}), M_b(q^S_{-1})] \\ &\supsetneq W^S[p_0, q_0], \\ M_b(W^U[p_{-1}, q_{-1}]) &= W^U[M_b(p^U_{-1}), M_b(q^U_{-1})] \\ &\subsetneq W^U[p_0, q_0], \\ M_b(W^S[q_{-1}, p_0]) &= W^S[M_b(q^S_{-1}), M_b(p^S_0)] \\ &\subsetneq W^S[q_0, p_1], \\ M_b(W^U[q_{-1}, p_0]) &= W^U[M_b(q^U_{-1}), M_b(p^U_0)] \\ &\supsetneq W^U[q_0, p_1]. \end{aligned} \tag{12}$$

D. Boundaries of the lobes E_0 and C_0

1. Concave pip ensures BIM boundary

We now show that constructing a transport boundary from BIM segments joined by a concave burning pip ensures that the E_0 and C_0 burning lobes are bounded by BIM segments. In reference to Fig. 11, we construct the transport boundary from $W^S[z^A, p_0]$ and $W^U[z^B, p_0]$ and the concave connection $\mathcal{C}$ (Fig. 10(b)).

This *TB* is the bounding front of *L*, and so by evolving *TB* forward, we can understand the boundary of the evolution of *L*. Because we chose a concave pip, $\partial\mathcal{M}(L) \subsetneq M_b(\partial L)$. We know by the invariance of BIMs and Fig. 11 that this transport boundary will map forward to $W^S[z^A, M_b(p^S_0)] \cup M_b(\mathcal{C}) \cup W^U[z^B, M_b(p^U_0)]$. We also know that the subsection of this front $W^S[p_1, M_b(p^S_0)] \cup M_b(\mathcal{C}) \cup W^U[p_1, M_b(p^U_0)]$ will lie behind the boundary of the burned region $W^S[z^A, p_1] \cup W^U[z^B, p_1]$ (see Fig. 12).

Recalling the set definition of the E_0 lobe (Eq. (7)), we see in Fig. 12 that $E_0 = \mathcal{M}(L) \cap R$ is bounded by $W^S[q_0, p_1]$ and $W^U[q_0, p_1]$. It is similarly straightforward to see that $C_0 = \mathcal{M}(L)^c \cap L$ is bounded by $W^S[p_0, q_0]$ and $W^U[p_0, q_0]$.

2. Convex pip creates non-BIM boundary segment

If a convex pip is chosen instead (p_0 in Fig. 13), the convex connection $\mathcal{C}$ used to join $W^S[z^A, p_0]$ and $W^U[z^B, p_0]$ leads to the following problem. The connection evolves into the arc $M_b(\mathcal{C})$ (Fig. 13 inset), which lags behind the continuation p_1. The space between $M_b(\mathcal{C})$ and the local stable and unstable segments near p_1 is part of L yet unburned, and therefore part of the C_0 lobe. Thus, the boundary of C_0 includes the arc $M_b(\mathcal{C})$, which is not a BIM segment. Therefore, in order to maintain the property that the E_0 and C_0 lobe boundaries are BIM segments, we must impose our second constraint. The active transport boundary must be defined by a *concave* burning pip.

E. Relations between fronts of preimages and preimages of fronts

The straightforward relation between burned region evolution $\mathcal{M}$ and bounding front propagation M_b has allowed us to understand the nature of the E_0 and C_0 lobes. Here, we establish an analogous relation between the loose and tight preimages (Eqs. (8)) of a region *A* and the bounding curves, so that we may similarly understand the E_{-1} and C_{-1} lobes. Figure 14 illustrates sets *A*, $\overline{\mathcal{M}}^{-1}(A)$ and $\underline{\mathcal{M}}^{-1}(A)$, as well as the relevant curves.

We begin with the loose preimage. Consider the set *D* of stimulations which do not reach the target set *A*, $D = \{p : \mathcal{M}(p) \cap A = \emptyset\} = \overline{\mathcal{M}}^{-1}(A)^c$; *D* is the white region in Figs. 14(b) and 14(e). The boundary of *D* maps into the boundary of *A* with orientation reversed, $M_b(\partial D) \subset \mathcal{R}(\partial A)$ (by reversal we mean $\mathcal{R}(\partial A) = \partial(A^c)$). Application of an inverse and second reversal yields $\mathcal{R}(\partial D) \subset \mathcal{R}M_b^{-1}\mathcal{R}(\partial A)$. Finally, since *D* is the complement of the set of interest, $\mathcal{R}(\partial D) = \partial\overline{\mathcal{M}}^{-1}(A)$. Thus, we have the following relation between the boundary of the loose preimage and *A* through the front element dynamics,

$$\partial\overline{\mathcal{M}}^{-1}(A) \subset \mathcal{R}M_b^{-1}\mathcal{R}(\partial A). \tag{13}$$

Equality is obtained when no local (swallowtails) or global front intersections develop in $\mathcal{R}M_b^{-1}\mathcal{R}(\partial A)$ (equivalently in $M_b^{-1}\mathcal{R}(\partial A)$). As we see comparing Figs. 14(b) and 14(e), a concave corner is certain to produce a swallowtail and break the equality.

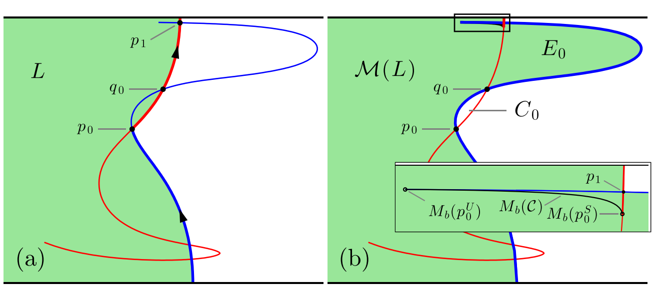

FIG. 12. (a) A concave pip is chosen to define the transport boundary. (b) *L* and *TB* are evolved forward showing the E_0 and C_0 lobes. Inset shows closeup of boxed region where $M_b(\mathcal{C})$ is within burned region. ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

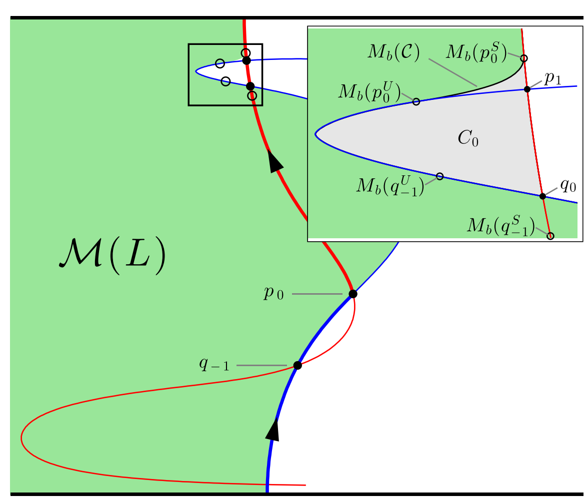

FIG. 13. Result of choosing convex pip for transport boundary (bold). Start with *L* burned, and iterate (filled green). Inset shows closeup of boxed region where $M_b(\mathcal{C})$ bounds burned from unburned giving C_0 lobe a non-BIM boundary. ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = 0$).

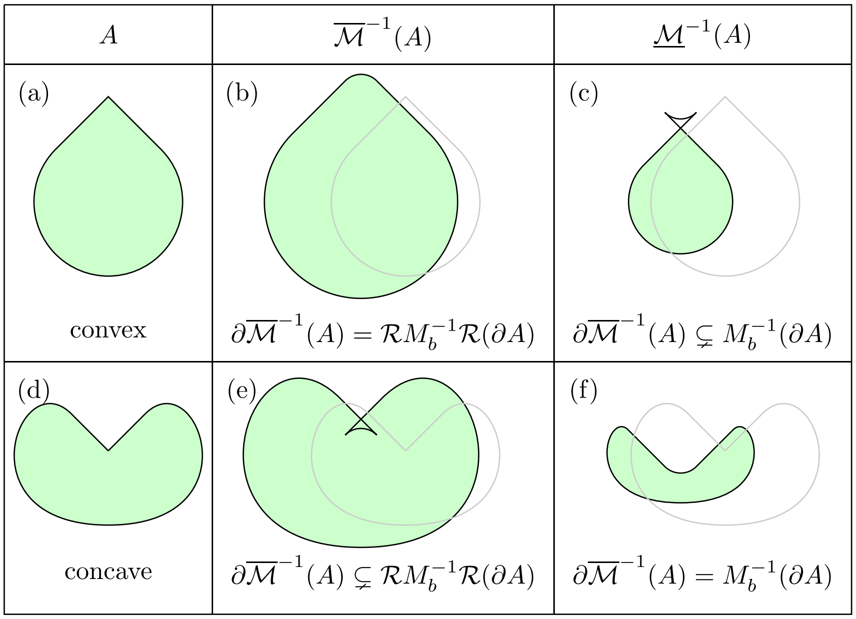

FIG. 14. Illustration of how convex (a) and concave (d) corners behave under loose ((b) and (e)) and tight ((c) and (f)) preimages. **u** is a uniform flow to the right. ((b), (c), (e), and (f)) Green regions are preimages of A. Black curves represent two ways of mapping ∂A backwards. Reference curves (gray) show original region.

The tight preimage relation is simpler to state

$$\partial\underline{\mathcal{M}}^{-1}(A) \subset M_b^{-1}(\partial A), \tag{14}$$

similarly with equality if no front intersections are developed in $M_b^{-1}(\partial A)$.

F. Boundaries of the lobes E_{-1} and C_{-1}

With the correspondence between bounding fronts and region preimages (Fig. 14) established, we can determine the E_{-1} and C_{-1} lobes with the assistance of the invariant manifolds. In Fig. 15(a), we define the *TB* using a concave pip.

The E_{-1} lobe depends on the loose preimage $\overline{\mathcal{M}}^{-1}(R)$ through Eq. (9). Let us examine this preimage. In Sec. VII E, we established the relation Eq. (13) between the boundary of a preimage and inverse images of boundaries. Here, this relation gives

$$\partial\overline{\mathcal{M}}^{-1}(R) \subset \mathcal{R}M_b^{-1}\mathcal{R}(\partial R) = \mathcal{R}M_b^{-1}(TB).$$

Figure 15(b) illustrates the inverse of the transport boundary $M_b^{-1}(TB)$ and the region it bounds (green). Since *TB* has a

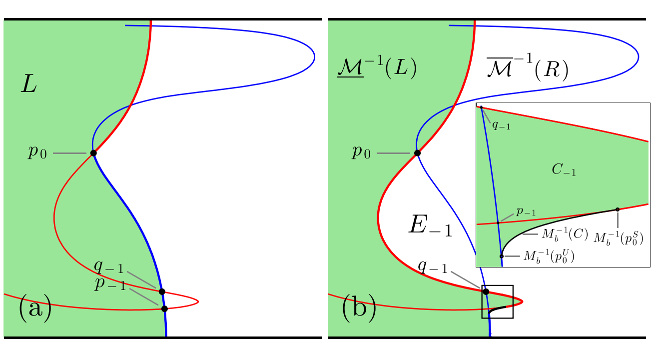

FIG. 15. (a) A concave pip is chosen to define the transport boundary. (b) R and TB are evolved backward (loose preimage) showing the E_{-1} and C_{-1} lobes. E_{-1} lobe is bounded by BIMs. Inset shows closeup of boxed region where $M_b^{-1}(\mathcal{C})$ bounds burned from unburned giving C_{-1} lobe a non-BIM boundary. ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

concave pip, $M_b^{-1}(TB)$ is not forced to generate a swallowtail. Our earlier assumption about v_0 being small enough to not generate any new intersections ensures that the entire curve $M_b^{-1}(TB)$ is free of self intersections. Thus, we have the equality $\partial\overline{\mathcal{M}}^{-1}(R) = \mathcal{R}M_b^{-1}(TB)$. This ensures that the loose preimage of R is exactly the white region in Fig. 15(b). Finally, intersecting with L (Eq. (9)), we see the location of E_{-1} and that its boundaries are the BIM segments $W^S[q_{-1}, p_0]$ and $W^U[q_{-1}, p_0]$.

On the other hand, C_{-1} depends on the tight preimage of the other side, $\underline{\mathcal{M}}^{-1}(L)$ (Eq. (9)). Equation (14) gives

$$\partial\underline{\mathcal{M}}^{-1}(L) \subset M_b^{-1}(\partial L) = M_b^{-1}(TB).$$

Therefore, $\underline{\mathcal{M}}^{-1}(L)$ is exactly the green shaded region in Fig. 15(b). Intersecting with R, we see that C_{-1} is not bounded by BIM segments $W^S[p_{-1}, q_{-1}]$ and $W^U[p_{-1}, q_{-1}]$ (see Fig. 15(b) inset). It is instead bounded by $W^S[M_b^{-1}(p_0^S), q_{-1}]$, $W^U[M_b^{-1}(p_0^U), q_{-1}]$, and $M_b^{-1}(\mathcal{C})$. Recall that we previously found that when a convex pip is used, the C_0 lobe is bounded in part by $M_b(\mathcal{C})$, a non-BIM segment.

If one is only interested in the escape process, i.e., transport of the burned region to the right of the *TB*, both concave and convex pips are equally suitable, in that E_{-1} and E_0 are both bounded entirely by BIMs. However, for the capture process, i.e., entrainment of unburned fluid to the left of the *TB*, either the lobe C_{-1} (for a concave pip) or the lobe C_0 (for a convex pip) will be bounded in part by a non-BIM segment. We shall preferentially consider concave pips in the remainder of this paper, both because the lobe C_0 is more directly visible in experiments and because the relevant *TB* (concave) can be realized through the dynamical development of a burned region.

G. Map inclusions

As can be seen in Figs. 16 and 17, the passive relations $C_0 = M_a(C_{-1})$ and $E_0 = M_a(E_{-1})$ are softened to the following inclusions:

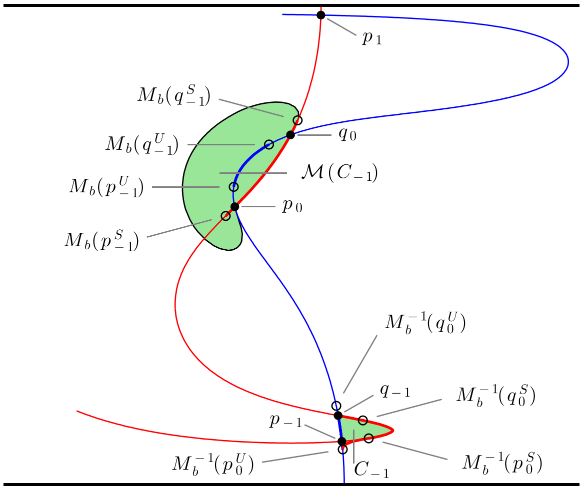

FIG. 16. Evolution of C_{-1} lobe and its bounding segments. ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

$$E_0 \subset \mathcal{M}(E_{-1}), \tag{15a}$$

$$C_0 \subset \mathcal{M}(C_{-1}). \tag{15b}$$

The E_0 inclusion follows directly from the definition Eq. (5). However, the C_0 inclusion depends on the fact that $M_b(W^S[p^S_{-1}, q^S_{-1}])$ includes $W^S[p_0, q_0]$, the stable boundary of C_0. We may reobtain E_0 by intersecting Eq. (15a) with R,

$$E_0 = \mathcal{M}(E_{-1}) \cap R. \tag{16}$$

VIII. ARBITRARY v_0: THE ROLE OF ADDITIONAL INTERSECTIONS

We have shown that for appropriately small v_0, there are new criteria that need to be taken into account when constructing a burning turnstile. These criteria arise from breaking the $v_0 = 0$ symmetry. In this section, we examine the case of larger v_0, where new intersections arise.

A. Swallowtails

In classical front propagation theory, formation of swallowtail catastrophes is generic[31] (Fig. 18). A front with a

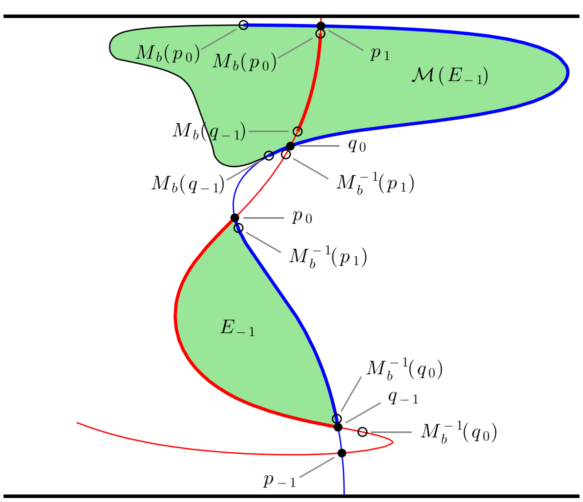

FIG. 17. Evolution of E_{-1} lobe and its bounding segments. ($v_0 = 0.05, b = 0.4, \omega = 5, \phi = \pi$).

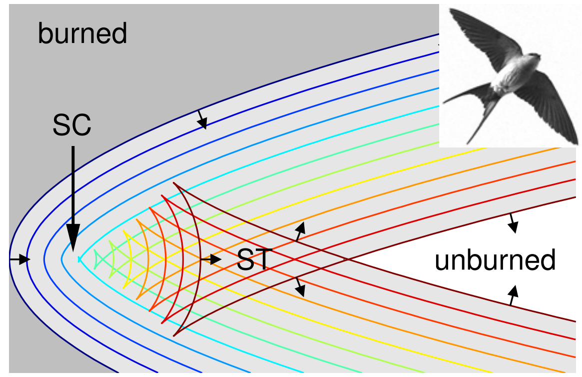

FIG. 18. The creation of a swallowtail in a system with no flow, as is common in standard optics and other front propagation phenomena. Burned region with parabolic boundary impinging from the left. Front elements first collide at a singularity—the swallowtail catastrophe (SC). The final swallowtail-shaped region (ST) has been traversed by two front segments. (inset: red-rumped swallow, wiki commons images).

concave segment has a point of greatest curvature or, equivalently, minimum radius of curvature. A swallowtail catastrophe occurs when this point propagates forward to its local center of curvature. As the front propagates further still, it develops a self-intersecting structure reminiscent of a swallow's tail. In the case of a front separating burned and unburned regions, the swallowtail must lie entirely within the burned region. That is, the front segments that make up the swallowtail are no longer boundaries between burned and unburned regions.

Swallowtail formation is also generic for fronts propagating in flowing media. The flow will influence the curvature of the front, thereby promoting or preventing formation of swallowtails. Many flows will tend to generate and regenerate areas of concavity in propagating fronts, thus facilitating continued swallowtail formation.

1. C_{-1} lobe swallowtail

Given the basic turnstile geometry in Fig. 4, we might expect to see swallowtails first appear in the boundary of the C_{-1} lobe.[32] This is because the lobe is long and narrow, creating a segment of high curvature at the tip. As the right-burning stable BIM deviates from the advective manifold (with increasing v_0), the BIM is forced inward, increasing the curvature at the tip until a swallowtail is forced.[31]

Let us consider the effect of such a swallowtail on the C_{-1} lobe. This lobe is defined as the set of all point stimulations that cross the transport boundary completely. Suppose a stimulation within the triangular region (point *a* in Fig. 19) was to evolve completely across the *TB*. Then, the last front element to pass through the stable BIM would be tangent to it and co-oriented. Hence, according to the no-passing lemma, this stimulation could not pass through entirely—a contradiction. The swallowtail region is, therefore, not a part of the C_{-1} lobe. The area remaining is shown in Fig. 19 (inset). The first effect of swallowtails, then, is that the C_{-1} area is diminished by the encroaching swallowtail.

The second effect is that the boundary of the C_{-1} lobe is reduced; here, the middle segment of the stable BIM is no longer on the lobe boundary (Fig. 19 inset). In general, a

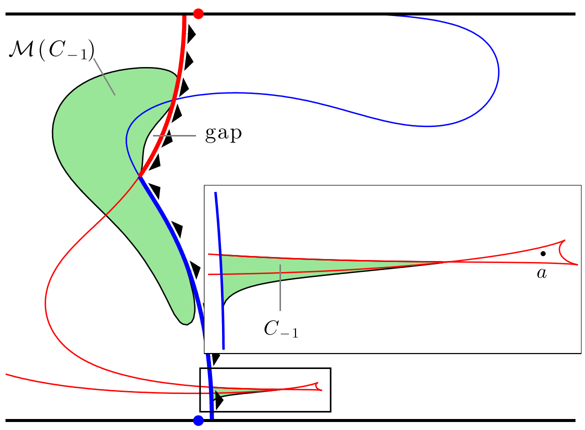

FIG. 19. Swallowtail diminishes size of C_{-1} lobe. Evolving C_{-1} forward covers most of C_0, but leaves a gap along the stable BIM. Inset shows closeup of boxed region where point a in swallowtail does not contribute to C_{-1}. ($v_0 = 0.085, b = 0.3, \omega = 5, \phi = \pi$).

single bounding BIM segment is replaced by the joining of two segments.

Evolving this reduced area forward, we find the situation in the upper-left of Fig. 19. The region bounded by stable and unstable BIMs is still the C_0 lobe—nothing in this region is reached by any stimulation in L. However, notice that a small portion also remains unburned by the evolution of C_{-1}. We can understand this by looking at Fig. 10(c); the intersection continuation runs ahead of the burned region leaving an unburned gap. In this case, the intersection continuation is lost when the swallowtail itself is removed by its catastrophe; at this point, we are left with the evolution of two distinct fronts (the boundary of the evolving C_{-1} lobe and the now unfolded segment of stable BIM). The arc that bounds the unburned gap in Fig. 10(c) is in Fig. 19 geometrically distorted by the flow, reversing its concavity.

Our previous inclusion relation for the evolution of C_{-1} (Eqs. (15b)) has thus been weakened to: if $C_{-1} \neq \emptyset$, then $C_0 \cap \mathcal{M}(C_{-1}) \neq \emptyset$.

2. *C_0 lobe swallowtail*

In the same spirit, the C_0 lobe may form a swallowtail as the unstable BIM deviates inward from the advective unstable manifold (Fig. 20). This swallowtail region represents fluid that has been crossed by two separate front segments of the evolving L. As C_0 is the region that is not burned by any stimulation in L, this swallowtail region is excluded from the C_0 lobe.

The unstable BIM segment that evolves to form this swallowtail previously bounded part of the C_{-1} lobe. However, note that this segment is oriented into C_{-1}. Therefore, evolving C_{-1} forward, the whole unstable BIM segment (whether it forms a swallowtail or not) is immediately burned through by $\mathcal{M}(C_{-1})$. Therefore, this swallowtail on the *unstable* manifold does not itself negate the inclusion relation $C_0 \subset \mathcal{M}(C_{-1})$.

Just as we saw in the C_{-1} lobe, the existence of a C_0 swallowtail only removes lobe boundary and so this boundary is still composed of BIM segments.

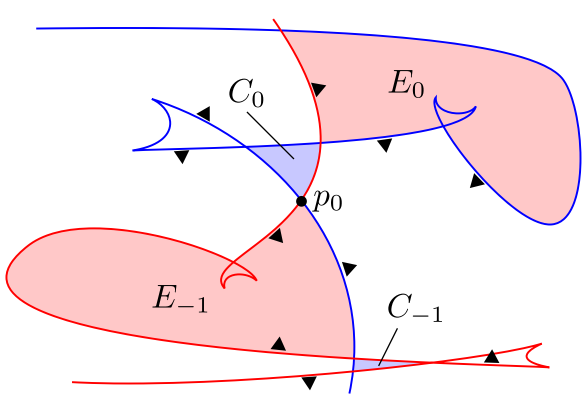

FIG. 20. Cartoon illustrating four basic swallowtails in the burning turnstile—one in each lobe. More complex flows can generate yet more swallowtails.

3. *E_{-1} and E_0 lobe swallowtails*

Adjusting the flow appropriately, the advective E_0 lobe becomes longer, thinner, and begins to fold. The active E_0 lobe can develop a swallowtail in the concave folding region (Fig. 20). This area is "burned twice" and so is still within the E_0 lobe. The inclusion relation remains $E_0 \subset \mathcal{M}(E_{-1})$.

The E_{-1} burning lobe can be similarly modified. As in the C_{-1} case, this swallowtail unfolds in forward time. However, since stimulations in E_{-1} need only intersect R, the swallowtail is included in E_{-1}. The inclusion relation is also unaffected.

B. Existence of pips

Unlike advective pips, which in our example flow (Eq. (1)) exist for any amount of periodic perturbation, it is possible for burning pips to be pulled apart in certain parameter regimes—for instance, when the front propagation speed is large enough. We can understand the basic intuition geometrically. Figure 21 shows BIMs for the same flow with four increasing values of v_0. In the left-most case ($v_0 = 0$), the BIMs degenerate to the advective invariant manifolds. All four turnstile lobes have the same area. This is because of area-preservation of the map (incompressibility of the fluid) and zero-net-flux across the turnstile. As v_0 is increased, the stable and unstable BIMs of interest are to the left and right of their respective advective fixed points. Also, the distance from each fixed point increases, squeezing the capture lobes to zero area, and finally destroying the central

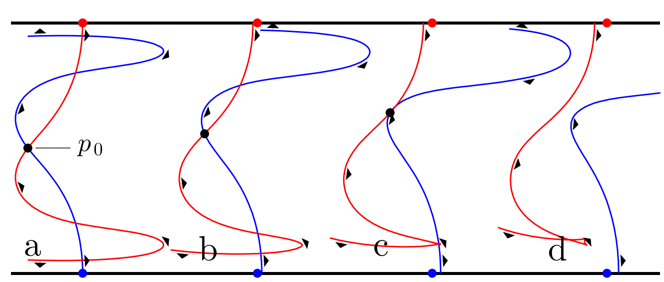

FIG. 21. Increasing $v_0 = [0, 0.037, 0.074, 0.1]$ from left to right. C_0 and C_{-1} lobes become empty and burning pips are lost. Dots show advective fixed points. ($b = 0.4, \omega = 5, \phi = \pi$).

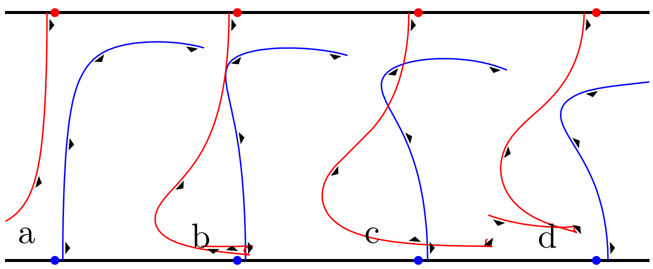

FIG. 22. Increasing $b = [0.133, 0.267, 0.400, 0.533]$, we see the finite gap caused by v_0 overcome by oscillations of the BIMs. Greater b does not necessarily correspond to larger lobes. Dots show advective fixed points. $(v_0 = 0.1, \omega = 5, \phi = \pi)$.

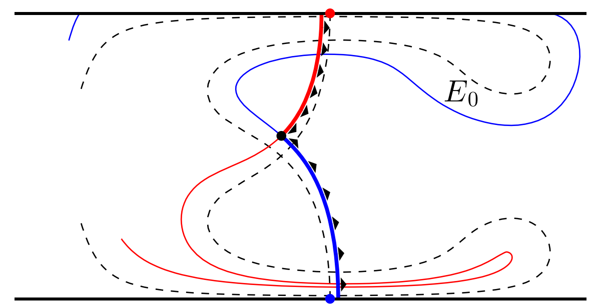

FIG. 23. The deviation of the BIMs from the advective manifolds (dashed) is sufficient that the unstable BIM runs into the channel wall. The channel wall thus forms part of the E_0 lobe boundary. $(v_0 = 0.075, b = 0.3, \omega = 4, \phi = \pi)$.

pip. Note that this loss of area is a mechanism distinct from the swallowtail discussed above.

It is also interesting to see the heteroclinic connection in Fig. 21(c). As previously discussed, intersections of fronts in xy-space are not generally intersections in $xy\theta$-space. In this case, since p_0 is a tangency between BIM projections, this is a true intersection of BIMs and thus p_0 generates a heteroclinic orbit.

We can understand the pip existence issue from another perspective. In the time-independent flow, the stable and unstable BIMs are separated by some finite gap (see Fig. 22(a) and Appendix for details). Sufficient forcing $(b > 0)$ causes oscillations in the BIMs on this same length scale—enough to generate intersections and lobes. The dependence of pip existence and lobe area on forcing strength appears to be more complex than on burning speed. For instance in Fig. 22, we see that the pips gained by some forcing in (c) are then lost by even greater forcing in (d).

In short, there are several distinct ways in which a system may gain or lose a burning pip. While the loss of this pip prevents the description of front propagation as mediated by burning lobes, it generically does not prevent fronts from propagating. As in passive advection, the turnstile is only one possible transport mechanism. Another mode of passive transport is ballistic transport, or "jets." Some open questions include: What other (non-lobe) mechanisms are available for propagating fronts? How can one decompose the evolution of a particular front into these various mechanisms?

C. Channel wall

In this paper, we have considered fluids within a channel geometry. A stimulation made in this channel with no flow will propagate outwards as a circle with radius growing at speed v_0 until it runs into the channel wall. A burned region that encounters the channel wall finds nothing beyond to burn, and so its bounding front stops progress in that direction.

While this is straightforward to understand, its implementation is worth noting. Numerically, we supply the channel with a thin boundary layer, extending the domain just beyond the channel wall. The particular flow $\mathbf{u}$ chosen beyond the boundary is not important. Upon entering this boundary layer, a front element's velocity perpendicular to the channel is quickly attenuated, reaching zero at the outer limit. In this way, front elements that enter the boundary layer do not leave.

In the ideal limit, the boundary layer is infinitely thin, and front elements that enter this layer can be identified with points on the channel wall. The result is that when lobes (such as the E_0 lobe in Fig. 23) collide with the channel wall, the channel wall itself forms a segment of the lobe boundary.

IX. APPLICATION: ALGAE INVASION REVISITED

As we have developed in this paper, the classical advective turnstile may be modified to account for the addition of propagating fronts. An active constituent in the ocean, such as plankton, would be transported into the bay by the advective turnstile while simultaneously growing outward, pushing past the advective invariant manifolds to the BIMs, which define the burning turnstile. This is why the advective lobe-based protocols (Fig. 1) were not effective.

We propose that the best strategy, given the assumptions, is to apply the algaecide to a burning escape lobe E_{-1} or E_0. This protocol eliminates exactly that algae which may transgress or has transgressed the transport boundary. Which of the two is the most efficient (smallest area) depends on the details of the flow.

Figure 24 illustrates these two new protocols. Both burning lobe-based protocols completely prevent the algae's invasion; both also use less suppressant than the fattened advective lobe—the previous best protocol considered in Fig. 1(d).

For advective transport, one can readily show that the *TB* defined by a pip connecting stable and unstable manifolds minimizes the flux across the *TB*. A small perturbation to such a *TB* results in an advective flux that is larger than (or equal to) the original flux. We conjecture that an analogous statement is true concerning the *TB* for front propagation. This question, however, is more subtle than in the passive case and will be left to a future publication.

X. SUMMARY

We demonstrate how the turnstile, a fundamental construct in the theory of passive fluid transport, is generalized to fronts propagating in a flowing fluid. This burning turnstile is originally defined in terms of the evolution of point stimulations in the active fluid. The modified turnstile

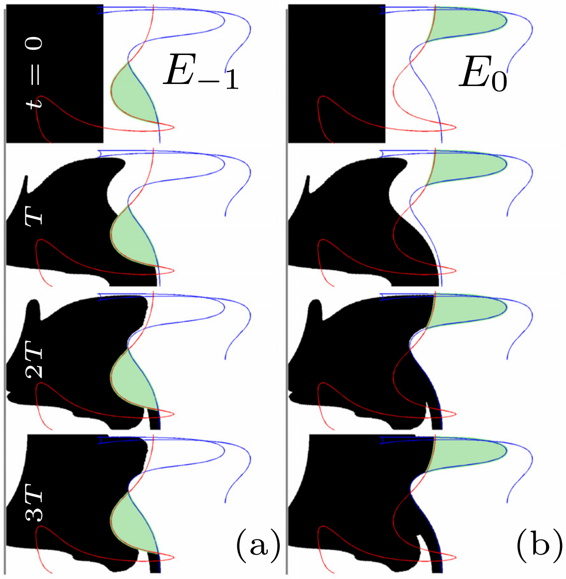

FIG. 24. As in Fig. 1, algae (black) impinges from left. Suppression using burning lobes (either E_{-1} or E_0) completely prevents invasion. Treated areas: $|E_{-1}| = 0.079$, $|E_0| = 0.093$.

exhibits several features distinct from its passive counterpart. First, the burning turnstile is bounded by BIMs, *one-sided* barriers to front propagation. Second, to form a meaningful transport boundary from finite segments of stable and unstable BIMs, the BIMs relative orientations must be consistent. Third, the burning pips used to join the stable and unstable BIMS are typically only intersections of the BIMs when projected to the *xy* plane. Fourth, this implies that a segment of the boundary of either C_{-1} or C_0 does not consist of BIM segments, depending on whether the pip is concave or convex, respectively. Finally, the areas of the lobes are not preserved and the relations between $\mathcal{M}(E_{-1})$ and E_0 and between $\mathcal{M}(C_{-1})$ are C_0 are more complicated than the advective case.

Increasing the front propagation speed can change the topology of the lobes, most notably by generically introducing swallowtails into the lobe boundaries. These swallowtails excise a portion of a lobe's BIM boundary. A large enough propagation speed can even remove a burning pip entirely. If no such pips remain, the turnstile formulation can no longer be applied and a more general theory is required.

To illustrate the utility of this theoretical framework, we apply it to a simple model of an ocean bay invaded by algae. This shows that efficient protection of the bay depends on the geometric constructs discussed herein—the burning lobes.

In addition to such control problems, there is significant interest in characterizing long-range mean propagation speeds in flowing systems. The above formalization of burning lobes allows us to now make simple but precise statements about long-range transport. For example, in a spatially periodic system such as the AVC, a long-range mean speed greater than one cell per forcing cycle requires that the E_0 lobe of one cell overlap with the E_{-1} lobe of the next cell. Preliminary studies also suggest the feasibility of a more refined lobe-dynamics-type approach.

ACKNOWLEDGMENTS

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APPENDIX: NO BURNING LOBES IN TIME-INDEPENDENT FLUID FLOW

It is well-known that there are no transverse intersections and hence no advective lobes in time-independent 2D flows. These flows also have no burning lobes, however the reasoning behind this is somewhat different. Assume for the remainder of this section that the fluid flow is time-independent. As previously shown,[10] BIMs obey the front-compatibility criterion; $\hat{\mathbf{g}} \equiv (\cos\theta, \sin\theta) \propto (dx/d\lambda, dy/d\lambda)$, where λ is a parameterization of the BIM by the Euclidean distance along the BIM. This is a condition on spatial structure. Additionally, BIMs are, by definition, time-invariant for time-independent flows. A front element that begins anywhere on the BIM travels in time along the BIM spatially—$(dx/dt, dy/dt) \propto (dx/d\lambda, dy/d\lambda)$. These two proportionalities combine to yield $\hat{\mathbf{g}} \equiv (\cos\theta, \sin\theta) \propto (dx/dt, dy/dt) = \mathbf{u} + v_0\hat{\mathbf{n}}$. This implies $\mathbf{u} \cdot \hat{\mathbf{n}} + v_0 = 0$. This is our main constraint on BIMs in time-independent flows. If $|\mathbf{u}| < v_0$, there are no solutions. If $|\mathbf{u}| = v_0$, there is one solution $\hat{\mathbf{n}} = -\mathbf{u}/v_0$. If $|\mathbf{u}| > v_0$, there are two solutions which we illustrate in Fig. 25. Consider the (hypothetical) intersection of two BIM segments at a burning pip. If at this intersection point there is one constraint solution, both BIM segments pass through the same $xy\theta$ point. By uniqueness of solutions to ODEs, these trajectories are the same and so there are no burning lobes. If there are two constraint solutions, the two BIM segments pass through the two different points (x_0, y_0, θ_0) and (x_0, y_0, θ_0'). If they pass through the same point, the previous argument applies again. Therefore, the intersection must be transverse and the two front elements involved must bear the relation in Fig. 25. Finally, notice that a burning pip, such as p_0 or q_0 in Fig. 11, does not have this configuration. Hence we reach a contradiction. This demonstrates that time-independent flows do not allow transverse pips or burning lobes.

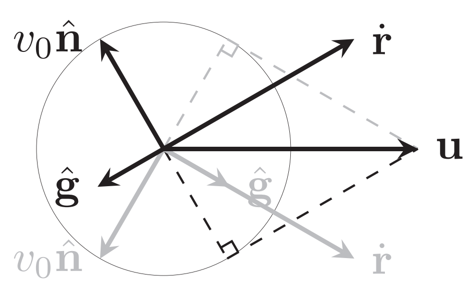

FIG. 25. For any fluid velocity vector $\mathbf{u}$ where $|\mathbf{u}| > v_0$, there are two solutions (orientations of $\hat{\mathbf{g}}, \hat{\mathbf{n}}$) to the constraint $\mathbf{u} \cdot \hat{\mathbf{n}} + v_0 = 0$. Black solution: $\dot{\mathbf{r}}$, $\hat{\mathbf{g}}$ antiparallel. Gray solution: $\dot{\mathbf{r}}$, $\hat{\mathbf{g}}$ parallel.

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